DUBLIN CITY UNIVERSITY

AUGUST/RESIT EXAMINATIONS 2015/2016

MODULE: CA320 - Computability and Complexity

PROGRAMME(S):

CASE - BSc in Computer Applications (Sft.Eng.)
CPSSD - BSc in ComputationalProblem SolvandSW Dev.

YEAR OF STUDY: 3

EXAMINERS: David Sinclair (Ph:6744)

TIME ALLOWED: 2 hours

INSTRUCTIONS: Answer 4 Questions.
At least 1 question must be from Section A.
All questions carry equal marks.

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The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

Requirements for this paper (Please mark (X) as appropriate)

☐	*Log Tables*	☐	*Thermodynamic Tables*
☐	*Graph Paper*	☐	*Actuarial Tables*
☐	*Dictionaries*	☐	*MCQ Only - Do not publish*
☐	*Statistical Tables*	☐	*Attached Answer Sheet*

Section A

QUESTION 1 ***[Total marks: 25]***

1(a) [5 Marks]

What are function types in Haskell? How is a function type written in Haskell? What is the implication of the particular form Haskell uses for defining a function type and invoking functions?

1(b) [8 Marks]

What are *class constraints* in Haskell? Give an example of how to use a *class constraint*.

1(c) [12 Marks]

Write a Haskell function, **without using the reverse function** , that tests whether or not a list is a "palindrome", i.e. the list is equal to its reverse. The function should use class constraints to ensure that it will work for lists of integers, floats, and characters.

[End Question 1]

QUESTION 2 ***[Total marks: 25]***

2(a) [5 Marks]

In the context of Haskell, describe what is meant by *lazy evaluation* and give an example of it.

2(b) [5 Marks]

Write a *data type* for a binary tree.

2(c) [15 Marks]

A reverse *monkey puzzle sort* of a list is performed by storing the elements of the list into a binary tree and then traversing the tree so that the nodes of a subtree are visited in the order right child, parent and then left child. Write a reverse monkey puzzle sort in Haskell.

[End Question 2]

Section B

QUESTION 3 ***[Total marks: 25]***

3(a) [5 Marks]

In the study of Computability & Complexity, why do we focus on languages?

3(b) [5 Marks]

Describe the terms *computability* and *complexity*. How do these terms relate to one another?

3(c) [15 Marks]

What is a *Universal Turing Machine*? What is its significance?

[End Question 3]

QUESTION 4 ***[Total marks: 25]***

4(a) [8 Marks]

In *computability* what does it mean when a problem is *undecidable*?

4(b) [10 Marks]

Prove that the *Halting Problem* is undecidable.

4(c) [7 Marks]

Describe how *reducibility* can be used to show a problem is undecidable.

[End Question 4]

QUESTION 5 ***[Total marks: 25]***

5(a) [5 Marks]

Describe the complexity classes $\mathbf{P}$ and $\mathbf{NP}$ and what is the relationship between them?

5(b) [10 Marks]

Prove that a language L belongs to $\mathbf{NP}$ if and only if a language $L_{check} \in \mathbf{P}$ and a polynomial p exist such that

$$L = \{x | \exists y . x, y \in L_{check} \wedge y \leq p(x)\}$$

5(c) [10 Marks]

Describe a problem that belongs to the class $\mathbf{NP}$ and, using the theorem in part(b) of this question, prove that it is a $\mathbf{NP}$ problem.

[End Question 5]

[END OF EXAM]